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MATH-ALIVE! USING ORIGINAL SOURCES TO TEACH MATHEMATICS IN SOCIAL CONTEXT

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ABSTRACT: There is a growing trend in mathematics education to use history and original sources to motivate students to learn content. An added benefit of using these sources is that it allows instructors to integrate a humanistic understanding of mathematics into a mathematics class by embedding the content within its motivating social context. This article describes how a simple question led to a fascinating quest through primary sources. Once collected, these sources proved to be a treasure trove of material illuminating mathematics and mathematicians in historical context.

KEYWORDS: History of mathematics, original sources, logistic equation, modelling, humanistic mathematics.

INTRODUCTION

There is a common perception of mathematics as a finished product invented by dead geniuses. In an effort to dispel this notion and convey to my students the excitement of mathematics as a living, breathing, and growing body of knowledge, created by human beings very much like ourselves, I have turned to original sources. A careful reading of these works can also reveal the social context within which mathematics is embedded. It is easier (and less threatening) to discern the influence of biases and hidden assumptions on the thinking of those from a previous era, than to question the underlying paradigms of one's own time. Once students are exposed to an historical case study where it is clear how self-interest and prevailing norms helped

determine what questions were asked, as well as how they were answered, they become more willing and able to evaluate critically and understand the social context as a component in the creation of all mathematics.

THE PROBLEM

How does one *find* original sources to use in the classroom? Often this happens serendipitously, as is illustrated by the following story of my search to find out how the logistic equation got its name. The *logistic equation*

$$y = \frac{be^{ax}}{1 + ce^{ax}}$$

is used to model natural systems, including human behavior, involving growth with limited resources. This simple equation, along with the differential equation it solves, and its familiar S-shaped curve, is ubiquitous, familiar to mathematicians, and natural and social scientists alike.

Certain I was not the first to wonder how this particular equation came to be called the "logistic," I began asking around. One colleague, in Biology, suggested that it had to do with the lodging of troops. Indeed, Manfred Schroeder writes "The designation logistic ... is derived from the French *logistique*, referring to the lodgment of troops." (emphasis in original) [9, p. 269]. Elsewhere, I read it simply meant "rational."[1] Both these explanations seemed plausible, but not conclusive. Then I serendiptously came across the following usage by Maria Agnesi,[2] an Italian mathematician from the eighteenth century, eldest of 21 children, who wrote one of the first calculus texts as a study guide for her younger brothers: "In these cases, therefore, we are obliged to have recourse to other methods. There are two of these which will assist us. One is, by means of a curve which is called the *Logarithmic Curve*, or the *Logistic*." (emphasis in original) [1, p. 12]. Hmmmm, I thought, perhaps logistic refers to "log-like" rather than lodgments or logic. I decided to try to trace back the usage through original sources. In the process of digging, I unearthed a gold mine of texts that went far beyond answering my question. First, I will address the question, then I will suggest a few of the many ways these materials can be used not only to teach the content (modelling, differential equations, etc.), but

[1] "The equation for the Verhulst model, if graphically plotted, gives rise to an S-shaped curve known as a logistic curve (which is from the Greek *logistikos*, meaning rational)" [16, p. 123].

[2] A student of mine was writing a thesis on Agnesi, using some of her original work.

also to impress upon students how new mathematics arises out of social, political, and economic, as well as intellectual needs.

THE SEARCH

The first place I looked was in the oft-cited classic paper by Pearl and Reed [7], who are usually credited with being the first to use the logistic equation to describe the growth of the population of the United States. A thorough reading revealed that although they do indeed introduce the equation as an appropriate model, nowhere do the authors actually call it "the logistic equation." In the endnotes to the chapter on ecological models in Michael Olinick's excellent text on modelling [5], I found a citation for G. Udny Yule's presidential address to the Royal Statistical Society in London [18] which contained complete references and a critical history of the logistic model. From Yule,[3] I learned that Pierre-Francois Verhulst (1804-1849), a Belgian sociologist and mathematician, was actually the first to propose and publish a formula for the law of growth for a population confined to a specified area [17]. Yule states, "[p]robably owing to the fact that Verhulst was greatly in advance of his time, and that the then existing data were quite inadequate to form any effective test of his views, his memoirs fell into oblivion." [18, p. 4]. Apparently, some eighty years later Pearl and Reed had arrived independently at the same result. Verhulst's work did eventually come to their attention, and in fact, Yule acknowledges that he is indebted to Pearl's book *The Biology of Death* [6] for the references to Verhulst. Yule goes on to say that Verhulst's memoirs "are classics on the subject, and as the originals are not very accessible, I have given some account of them in Appendix I." [18, p. 4]. Excitedly, I turned to the appendix where Yule carefully explains Verhulst's derivation of the formula, only to find the tantalizing hint: "Verhulst now christens the curve a 'logistic,'" [18, p. 44] with no further explanation as to why. So I continued my quest by searching for the "not very accessible" original paper by Verhulst.

Several months later, after consulting with our reference librarian (who eagerly took up the challenge), I received a note informing me that the document was available on microfilm through interlibrary loan. Verhulst wrote in French, but with dictionary in hand and my rudimentary high school background, I was able to comprehend the paper. On page 8, my efforts were rewarded, and I found what I was looking for. Here is the evidence; judge for yourself.

[3] Olinick is also fully aware of Verhulst's contributions and includes a biography of Verhulst in his text.

THE QUESTION ANSWERED

The logistic equation is usually written with population expressed as a function of time (population is the dependent variable). This familiar form of the equation involves an exponential function. Verhulst wrote the relationship with time as the dependent variable.[4] Since the logarithmic function is the inverse of the exponential, his equation had t (time) equal to a (somewhat complicated) logarithmic function of p (population):

$$t = (1/m)\log\left[\frac{p(m-nb)}{b(m-np)}\right]$$

In the next sentence, Verhulst indeed "christens" the equation: "We give the name logistic to the curve (see Figure 1) characterized by the preceding equation."[5] [17, p. 8]. The diagram clinched it for me: there two curves labelled "Logistique" and "Logarithmique" are drawn on the same axes, and one can see that there is a region where they match almost exactly, and then diverge.

I concluded that Verhulst's intention in naming the curve was indeed to suggest this comparison, and that "logistic" was meant to convey the curve's "log-like" quality.

As an unanticipated bonus, along with satisfying my curiosity, I now had in my possession a rich collection of original sources (the papers of Verhulst, Pearl and Reed, and Yule). I have used these materials to teach content[6] to students in calculus, differential equations and modelling, and in a summer institute for K-12 teachers. I also assign readings from these papers that lead naturally to discussions of mathematics in social context. Here I give you a sample of some of the issues that arise. My goal is not to provide specific lessons, but rather to suggest by example how to integrate a humanistic understanding of mathematics into a mathematics class by embedding the content within its motivating social context.

MATHEMATICS IN SOCIAL CONTEXT

Pearl and Reed begin their paper by explaining why "mathematical expressions of population growth [that are] purely empirical... have a distinct and considerable usefulness." [7, p. 275]. The census is only taken every ten

[4] The differential equation model he derived and solved was $dt = (M\,dp)/(p(m-np))$.

[5] "Nous donnerons le nom de *logistique* à la courbe (voyez la figure) caractérisée par l'équation précédente" [17, p. 8].

[6] See [10].

years and it is often necessary to get an accurate estimate of the population in inter-censal years (the years following the last census, as well as the years lying between prior censuses). "For purposes of practical statistics it is highly important to have these inter-censal estimates of population as accurate as possible, particularly for the use of the vital statistician, who must have these figures for the calculation of annual death rates, birth rates, and the like." [7, p. 275]. We will return to the uses to which these figures are put, as well as the important social issues affecting even this seemingly straightforward calculation of birth and death rates.

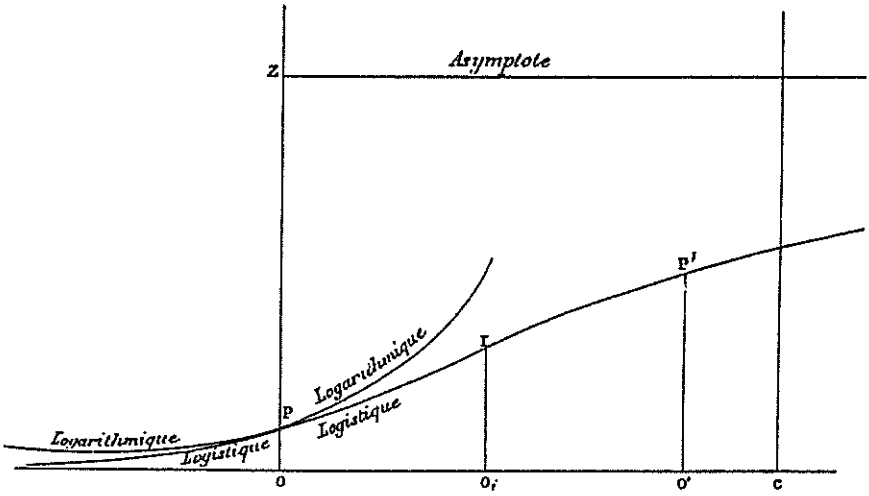

Figure 1. Verhulst's diagram after he christens the curve logistic. [17, between pages 15 and 16.]

The drawback of a purely empirical expression is that it does not give a general law of population growth: "No process of empirically graduating raw data with a curve can in and of itself demonstrate the fundamental law which causes the occurring change." [7, p. 275]. Dissatisfied with existing empirical representations of population growth in the United States, even when their accuracy is "entirely sufficient for all practical statistical purposes" [7, p. 280], Pearl and Reed find it "worthwhile to attempt to develop such a law [of population growth], first by formulating a hypothesis which rigorously meets the logical requirements, and then by seeing whether in fact the hypothesis fits the known facts." [7, p. 281]. So they list a set of conditions that should be fulfilled by any equation meant to describe the

growth of population in an area of fixed limits. These are the explicit underlying assumptions of the logistic model. They are reasonable enough: we assume

- the population is always increasing,
- when resources are plentiful the rate of increase is itself increasing, and
- as resources become more scarce, the rate of increase begins to decrease, until the population approaches the carrying capacity of the restricted area.

One equation which fulfills these requirements is

$$y = \frac{be^{ax}}{1 + ce^{ax}}$$

which is equivalent to the equation Verhulst named the logistic. Pearl and Reed then demonstrate that this equation gives as good a fit to the observations as the "logarithmic parabola,"[7] the curve which "describes the changes which have occurred in the population of the United States in respect of its gross magnitude, with a higher degree of accuracy than any empirical formula hitherto applied to the purpose." [7, p. 280].

This result is significant because "[a logistic] curve which on a priori grounds meets the conditions which must be satisfied by a true law of population growth, actually describes with a substantial degree of accuracy what is now known of the population history of this country." [7, p. 284].

In this success lies a danger. It is a very human tendency to conflate our symbols with the things they represent; to confuse our models, which are merely our best attempts to make sense of the welter of phenomena we observe, with "laws" of nature – rules that phenomena must follow. The statisticians of the time were aware of these pitfalls. In response to a paper delivered by Dr. Stevenson, in the meeting following Yule's Presidential address, a Professor A. L. Bowley said: "I regret that so much prominence has been given to the logistic equation. It certainly has the merit, and the danger, of mathematical neatness...." [13, p. 76]. He warns that although "the logistic equation is well adapted to represent rather roughly the recorded changes of population in selected countries," he "fear[s] that, after the considerable advertisement it has received, many erroneous conclusions will be based on it." [13, p. 80]. We tend to enshrine the logistic model as the logistic law of population growth. This is not something we do consciously, but the way the subject is traditionally taught often leaves students with

[7] A quadratic equation with a logarithmic term added on, $y = a + bx + cx^2 + d\log x$.

this idea. When the subject matter is presented in the context in which it was created, students are properly warned of this danger.

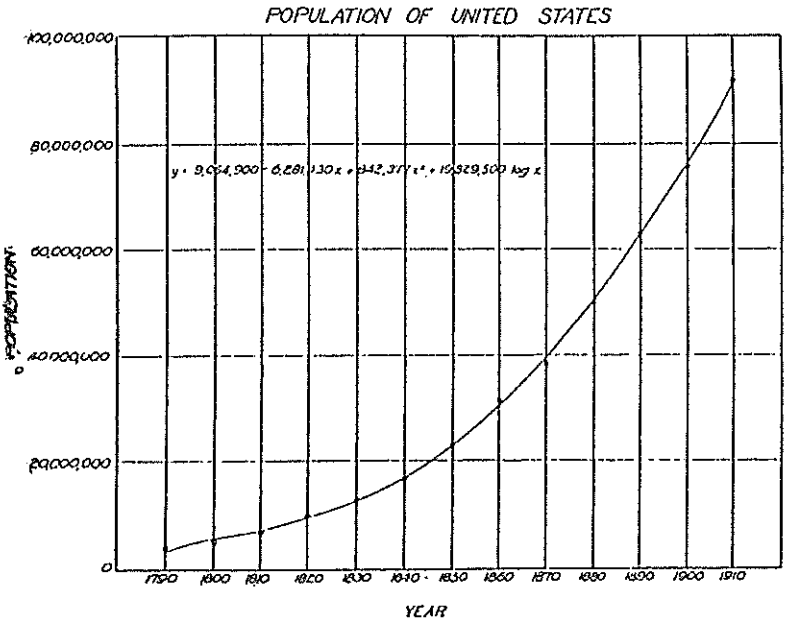

Figure 2. "Diagram showing observed and calculated populations (from logarithmic parabola) from 1790 to 1920."[7, p. 289]

Other mathematical models, based on different but equally reasonable assumptions, also resulted in curves that fit the data. Dr. Brownlee, another discussant of Dr. Stevenson's paper, "had been accustomed to graduate the populations in the country from 1801 onwards to a curve of a different form

$$y = Ae^{-\left(\frac{B}{C}e\right)^{Cx}}$$

a curve which developed from the idea that a sudden impulse to increase of population took place for some reason or other and then slackened in a geometrical progression." [13, p. 82]. Why was the logistic model the most popular mathematical model? And if other expressions, determined solely by curve-fitting techniques (and not based on mathematical models of assumptions about how populations should behave) yield curves that

empirically fit the data equally well or better,[8] why were statisticians so anxious to embrace a mathematical model?

THE REST OF THE STORY

There are good mathematical reasons that explain each of these questions, but they do not provide the whole story. Part of the context in which mathematics takes place includes social relations and societal values. In the early part of the twentieth century, Statistics, as a discipline, was still struggling to attain the status of a "science." In our society, Mathematics, the "queen of the sciences" confers respectability and higher status to any discipline that can frame its arguments in the language of equations and formulae. The presidential address by Yule set the Society firmly on the path of greater mathematization. This made many members uneasy. "Some apprehension has been expressed lest the highly technical trend taken by statistical investigation during the past thirty years may not have kept some from joining the Society, in consideration of their want of acquaintance with the higher mathematics." [18, p. 60]. Nevertheless, a mathematical model is to be preferred over a purely empirical one in part because it is mathematical.

And why did the logistic win out? Partly because its champion was President Yule. Professor Edgeworth says, "Altogether, on the question which I have raised concerning the credentials of the logistic law, I am disposed to defer much to what I should call Mr. Yule's 'authority.' ... He is conversant with the canons of evidence which satisfy a society dealing with sciences more rigid than statistics." [18, p. 59]. Scholars in science studies maintain that when several competing theories or models exist, other (social) factors, along with mathematical and scientific plausibility and reasonableness, help determine which theory or model prevails.[9] I believe we have here some evidence in support of this controversial claim.

OTHER INTERESTING DISCOVERIES

Let us now look more closely at some of the other assumptions hidden within this modelling process. We can ask, does the census really include everyone? Our logistic model may be a very good fit to only certain segments

[8] "The most that can be asserted is that [the logistic] equation gives nearly or quite as good a fit to the observations as does the logarithmic parabola." [7, p. 284].

[9] "[W]hether an account is powerful or not depends not simply on the force of the argument but also on the positioning and resources of its advocates....[W]e need to ask why particular readings hold such sway and others disappear without a trace." [4, pp. 352-3].

of the population. There are many illuminating examples throughout Yule's address (as well as in Verhulst's paper) of assumptions made about (and encoded into the models of) populations. Two examples from Yule are whether to take into account (or not) catastrophes like World War I and the influenza epidemic of 1918-19; and whether or not to include Alsace-Lorraine in the population of France. We read in Verhulst[10] that "We haven't taken account of immigration, because we have observed that it has been offset by the impediments that slavery brings to the growth in the number of blacks in the South."[11] [17, p. 5].

As many feminists have pointed out, categories create our reality.[12] In particular, census categories determine what is visible – what is counted as work, who is defined as a worker, how (or even if) class is measured, and so on. The census feeds almost every other defining and decision-making process in the country.[13] In fact, the United States census purposefully started out to include only a percentage of slaves as property and to exclude Native Americans.[14] Again, the logistic model may not fit the actual population.

Pearl and Reed explore some of the "consequences which flow from [the logistic] equation..." and make the claim that "the United States has already passed its period of most rapid population growth." Although they caution that "prophecy is a dangerous pastime," they go on to say that according to their model, "the maximum population which the continental United States, as now really limited, will ever have will be roughly twice the present population" (i.e., it will level off at just under 200 million). [7, pp. 284-6].

To support this claim, Pearl and Reed use mathematical concepts no more sophisticated than ratio and proportion. These few pages provide a fascinating case study for students to examine and critically evaluate quantitative arguments made by "experts" from a previous era (and they can check the validity of the predictions). To encourage them to apply the same skills to arguments made by the pundits of their own time, I turn to the pages of local newspapers and national magazines for examples in current events. The key point to emphasize and discuss is that important

[10] Verhulst uses population data from the United States, 1790-1840, as an example of a population growing in geometric progression (exponential growth).

[11] "Nous n'avons pas tenu compte des immigrations, parce que nous les regardons comme largement compensées par les obstacles que l'esclavage apporte à la multiplication des noirs, dans les États du Sud." [17, p. 5].

[12] For instance, see Gloria Steinem in [12, pp. 225-6].

[13] For more information about early U. S. censuses and their social inaccuracy, see [3].

[14] See [12] and references therein.

policy decisions are made based on the predictions arising from models.

Each meeting of the Royal Statistical Society began with a member presenting a paper, and then a discussion followed. Fortunately, the Proceedings include transcriptions of the discussions and thereby provide a treasure trove of examples of mathematics in a social context. For instance, in reply to Yule's Presidential address, Professor Edgeworth says,

> So far as the logistic law is regarded as a law of nature based on a *vera causa*, it confirms the suggestion of our President's [Yule's] important brochure in the *Fall of the Birth-Rate* (1920); to the effect that the retardation of population is likely to be more automatic and impersonal than some followers of Malthus represent; that we may be taking alarm too soon about the increase of population. Future generations may look back upon our fears that the birth-rate is too high or too low, much as we regard the alarms caused in the past by the rise or fall of prices....The inflation consequent on the war may, indeed, serve to illustrate the real dangers which may be threatened by excessive population in some Eastern countries. But in the West for the present it is suggested that alarm is premature. Posterity will perhaps smile at our solicitude." [18, pp. 59-60].

We are the aforementioned posterity – are we smiling? Here is another opportunity to learn from history. I ask students if they see any similarity between these arguments and responses to predictions about global warming, and other environmental disasters today.

The next meeting, after Yule's address, took place on December 16, 1924, and opened with T. H. C. Stevenson of the General Register Office presenting a paper on the general subject of "The Laws Governing Population." Stevenson, it becomes clear, is a physician. Much discussion in both meetings is devoted to the inter-relations of the birth- and death-rates. Clearly it is the relationship between them that determines the rate of population growth (or decline).[15] Addressing Yule, Stevenson says "I should like here, sir, to refer to your fine testimonial to my profession in attributing the decline of the death-rate, notwithstanding increased density of population, to 'the noble science of medicine,' and to thank you for those words." [13, p. 75]. He continues by saying that surely it is easier to prevent births than to prevent deaths, and that "civilized man" has been quite successful

[15] "[I]t is neither the death-rate alone nor the birth-rate alone which determines the natural rate of growth of a population, but the difference between the two." [18, p. 30].

in the more difficult task of preventing deaths, but he admits that he (civilized man) "may at the same time be attaining success also in the easier one, through the *less noble art* of contraception." (Emphasis my own.) [13, p. 75]. Imagine my surprise (and delight) to find that this remark does not go unnoticed, but is firmly challenged by a certain Dr. Marie Stopes. At last we get to hear a female voice making a contribution to the discussion. Addressing Dr. Stevenson, she said[16] that she felt he "was a little too ready in taking a halo unto himself. Why was it a nobler profession to deal with disease than to deal with the creation of a glorious thing like life? Was not prevention better than cure?" [13, p. 86].

Stopes notes that a previous speaker, Sir William Beveridge, had asked for some facts about the correlation between birth-rate and death-rate, and "as she was at present analysing some 5,000 cases at the first Birth Control Clinic, she had certain facts to give which no statistician had hitherto been able to obtain." [13, p. 85.] She goes on to explain that in looking at the correlation between birth- and death-rate, she had taken into account the pregnancy rate, "which was a social factor which seemed to her to matter more to the vitality of the stock than the actual birth-rate." Her results indicated that "where women were in a poor condition of ignorance, and housing conditions were bad, they tended to have an incredibly high pregnancy rate as distinct from the birth-rate." That is, the death-rate of a woman's pregnancies rose rapidly with the number of them.

In her comments on factors affecting birth- and death-rates, this intrepid female voice in the Royal Statistical Society[17] addresses an earlier remark by Dr. Brownlee which assumed that birth control did not count in 1871. "It would be better if Dr. Brownlee would read history; to say that there was no birth control at that date was nonsense. There was an enormous circulation of a book by Dr. Trall, published in 1866, of which 40,000 copies were sold, and four or five other publications were also being sold at that time, to say nothing of Francis Place, who in 1822 had leaflets which were circulated broadcast." Her final barb to Brownlee (I wish I could have seen the expression on the faces of her listeners) was that she had been "taught at school that mathematics were useful, but that truth must be put in at one end of the mill if truth were to come out at the other." [13, p. 86].

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[16] Dr. Stopes' remarks are reported in the third person, rather than being transcribed verbatim. When the discussant has prepared remarks and a text is available, they are quoted in the first person. The subsequent spontaneous responses are reported in the third person.

[17] Was she the only female member? Readers (and their students) are invited to do research and find out!

CONCLUSION

In reading the discussions included in the Proceedings, one gets a real sense of the personalities of the human beings wrestling with difficult social issues surrounding the mathematical model they are creating. These experts of their day (dead geniuses in ours) have a sense of humor. Consider this quip by Sir Athelstane Baines. (He is referring to Professor Pearl's experiments with fruit flies (Drosophila) which suggest that the death rate tends to increase with increasing density of population. [18, p. 24.]) "Finally, so far as the habits of the Drosophila are concerned, I must confess that I sympathize with Dickens' Eugene Wrayburn, who, when taunted with the example of the ant and the bee, 'protested, on principle, as a biped.'" [18, p.61] As we have seen above, they also experience math anxiety, and can be racist in their presuppositions. In other words, they are human beings, just like us.

These are just a few of the gems that can be found in original sources. I encourage you to go hunting and find your own treasures. Do not be intimidated. I have found that historical papers are often easier to read than their descendants in modern journals. In particular, the writing style of the mathematicians and physicists of the nineteenth and early twentieth century is less dry and more engaging, liberally sprinkled with charming and insightful metaphors. To help you get started, the list of references that follows includes several standard source books [2, 11, 14, 15] as well as lessons for teaching content using original sources in various classes [8, 10]. I have made suggestions throughout this article as to how one might use these materials to stimulate discussion about mathematics in social context. And, fully aware of the dangers of extrapolating from a sample of one, I nevertheless predict that your students will be as excited and motivated by this heady mix of content and context as my students have been. Happy hunting!

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BIOGRAPHICAL SKETCH

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